CAMBRIDGE
International Education

# Cambridge O Level

**CHEMISTRY** **5070/22**

Paper 2 Theory **October/November 2024**

MARK SCHEME

Maximum Mark: 80

**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.



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**PUBLISHED**

## Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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|---|
| GENERIC MARKING PRINCIPLE 5:<br><br>Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen). |
| GENERIC MARKING PRINCIPLE 6:<br><br>Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind. |

**Science-Specific Marking Principles**

| | |
|---|---|
| 1 | Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly. |
| 2 | The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored. |
| 3 | Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection). |
| 4 | The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted. |
| 5 | ‘List rule’ guidance<br><br>For questions that require ***n*** responses (e.g. State **two** reasons …):<br><br>• The response should be read as continuous prose, even when numbered answer spaces are provided.<br>• Any response marked *ignore* in the mark scheme should not count towards ***n***.<br>• Incorrect responses should not be awarded credit but will still count towards ***n***.<br>• Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.<br>• Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science. |

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6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient ($a$) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

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| Question | Answer | Marks |
|---|---|---|
| 1(a)(i) | **B** | **1** |
| 1(a)(ii) | **E** | **1** |
| 1(a)(iii) | **D** | **1** |
| 1(a)(iv) | **E** | **1** |
| 1(a)(v) | **C** | **1** |
| 1(b) | number of protons: 23 (1)<br><br>number of neutrons: 28 (1) | **2** |

| Question | Answer | Marks |
|---|---|---|
| 2(a)(i) | to produce heat / to heat the furnace | **1** |
| 2(a)(ii) | toxic / poisonous | **1** |
| 2(a)(iii) | $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$ | **1** |
| 2(b) | calcium carbonate decomposition to calcium oxide / calcium carbonate forms calcium oxide<br>$CaCO_3 \rightarrow CO_2 + CaO$ (1)<br><br>calcium oxide combines with impurities or sand or silicon (IV) oxide (in iron ore) /<br>$CaO + SiO_2 \rightarrow CaSiO_3$ (1) | **2** |

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| Question | Answer | Marks |
|---|---|---|
| 2(c) | **Any two from:**<br>have high density / more dense (1)<br>have coloured compounds (1)<br>variable oxidation numbers (1)<br>act as a catalyst (1)<br>not as reactive (1)<br>hard (1) | **2** |
| 2(d) | **Any three from:**<br>**M1** zinc forms a **protective** layer / zinc forms a barrier (1)<br>**M2** (barrier) to prevent oxygen reaching (surface of) iron / to prevent water reaching (surface of) iron (1)<br>**M3** zinc more reactive (than iron) (1)<br>**M4** zinc corrodes in preference (to iron) (1) | **3** |
| 2(e) | no effect (1)<br>same (number of) moles of **gas** on each side of the equation / equal volumes of **gas** on each side of equation (1) | **2** |
| 2(f) | $Fe_3O_4 + 8HCl \rightarrow FeCl_2 + 2FeCl_3 + 4H_2O$<br>correct formulae (1)<br>correct balancing (1) | **2** |

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| Question | Answer | Marks |
|---|---|---|
| 3(a) | decomposition of an aqueous ionic compound / decomposition of a molten ionic compound (1)<br>by electricity / (electric) current (1) | **2** |
| 3(b) | left hand electrode labelled | **1** |
| 3(c)(i) | hydrogen | **1** |
| 3(c)(ii) | $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$ | **1** |
| 3d | platinum | **1** |

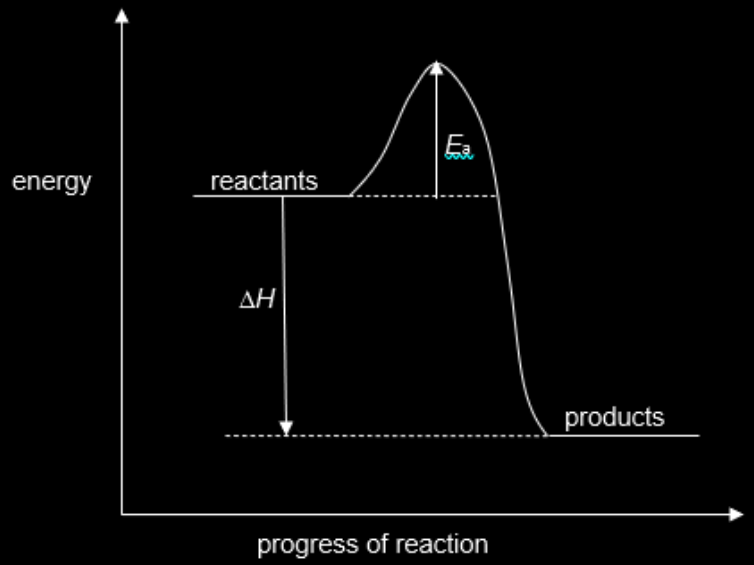
| Question | Answer | Marks |
|---|---|---|
| 4(a) | **Any two from:**<br>differ from one member to the next by $-CH_2-$ unit (1)<br>trend in physical properties (1)<br>similar chemical properties (1) | **2** |
| 4(b)(i) | **all** the carbon-carbon bonds are single bonds | **1** |
| 4(b)(ii) | $CH_3CH_2CH_2CH_3$ | **1** |
| 4(c)(i) | (chemical) feedstock | **1** |
| 4(c)(ii) | $C_9H_{20} \rightarrow C_3H_8 + 3C_2H_4$<br>correct formulae (1)<br>correct balancing (1) | **2** |
| 4(d)(i) | substitution | **1** |

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| Question | Answer | Marks |
|---|---|---|
| 4(d)(ii) | provides the activation energy (for the reaction) | **1** |
| 4(d)(iii) | energy associated with bond breaking 413 + 243 **OR** 656 (1)<br>energy associated with bond forming = 346 + 432 **OR** 778 (1)<br>(enthalpy change = +656 – 778) = –122 (kJ / mol) (1) | **3** |
| 4(e) | (only) a single product is formed | **1** |

| Question | Answer | Marks |
|---|---|---|
| 5(a) | $ZnCl_2(aq) + H_2(g)$ | **1** |
| 5(b)(i) | the gradient decreases (as time increases) | **1** |
| 5(b)(ii) | **M1** the particles get further apart (1)<br>**M2** frequency of collisions (of the particles) decreases (1) | **2** |
| 5(b)(iii) | rate increases<br>**and**<br>**M1** with powder because surface area greater (1)<br>**M2** collision frequency increases / more collisions per unit time / collisions more often / greater collision rate (1) | **2** |

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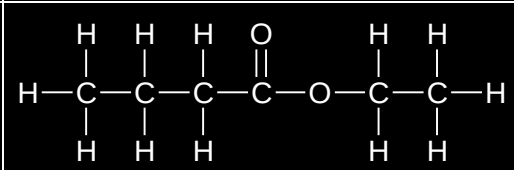
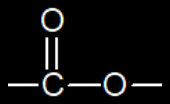
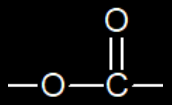
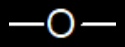
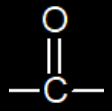
| Question | Answer | Marks |
|---|---|---|
| 5(c) | moles $HCl = 0.400 \times \frac{16.0}{1000}$ **OR** $6.4 \times 10^{-3}$ (1)<br><br>moles $H_2 = 3.2 \times 10^{-3}$ **OR** $\frac{6.4 \times 10^{-3}}{2}$ (1)<br><br>volume of hydrogen = 0.077 ($dm^3$) (1) | **3** |
| 5(d) |  <br><br>**M1** reactants to left and products to right and reactant level above product level (1)<br><br>**M2** activation energy hump between reactants and products and almost vertical upward arrow from reactant energy level to top of energy hump labelled activation energy (1)<br><br>**M3** enthalpy change shown between reactants and products as downward arrow labelled enthalpy change (1) | **3** |

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| Question | Answer | Marks |
|---|---|---|
| 5(e) | **M1** white precipitate (1)<br>**M2** dissolves / forms colourless solution (1) | **2** |
| 5(f) | **M1** filter (off excess zinc from aqueous zinc chloride) (1)<br>**M2** evaporate (filtrate) to crystallisation point / heat until saturated solution formed (1)<br>**M3** cool or leave (saturated solution), (filter) and dry (crystals) with filter paper (1) | **3** |

| Question | Answer | Marks |
|---|---|---|
| 6(a) | **M1** it is a giant (structure) (1)<br>**M2** the bonds are strong (1) | **2** |
| 6(b) | there are no mobile electrons | **1** |
| 6(c) | one pair of electrons in each overlap area and one lone pair on each nitrogen | **1** |
| 6(d)(i) | –3 | **1** |
| 6(d)(ii) | no change in (any of) the oxidation number | **1** |
| 6(e) | moles of $B_2O_3 = \frac{8.0}{70}$ **OR** 0.1143 (1)<br>moles of Mg $= \frac{7.2}{24}$ **OR** 0.3 (1)<br>$B_2O_3$ needs 0.3429 mol of Mg / Mg can only react with 0.1 mol of $B_2O_3$ (1) | **3** |

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| Question | Answer | Marks |
|---|---|---|
| 7(a)(i) | general formula | **1** |
| 7(a)(ii) | 5 | **1** |
| 7(b) |  | **1** |
| 7(c)(i) |  **M1** and / or  link between all 4 boxes (1)<br>**M2** ester linkages in correct directions (1)<br>**M3** left hand box ends  <br>**and**<br>right hand box ends in  (1) | **3** |
| 7(c)(ii) | condensation | **1** |
| 7(d) | **M1** PET broken down into monomers (1)<br>**M2** (monomers) re-polymerised (1) | **2** |
| 7(e)(i) | sodium ethanoate (1)<br>water (1)<br>carbon dioxide (1) | **3** |

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| Question | Answer | Marks |
|---|---|---|
| 7(e)(ii) | arrangement: irregular (1)<br><br>motion: sliding over each other (1) | **2** |

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